# The Impact of Mating on Thigmotaxis and Centrophobism in *Drosophila melanogaster*

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**Abstract (One paragraph format)**
In Drosophila melanogaster, mating behaviors are influenced by genetic factors and environmental cues, affecting social interactions and evolutionary outcomes. Taxis behaviors guide navigation and survival by responding to environmental stimuli. Centrophobism is a preference for peripheral locations over central spaces in an environment, reflecting innate anxiety-like behaviors and survival strategies. However, the impacts of mating on thigmotaxis and centrophobism behaviors remains largely unexplored. The research shows that *Drosophila* melanogaster exhibits a significant decrease in time spent in the center during mating compared to non-mating periods. Previously, it was thought that behaviors in *Drosophila* melanogaster might not significantly alter spatial preferences within an environment, focusing instead on the role of external stimuli such as pheromones and visual cues. The result of this study adds to our understanding by demonstrating a clear shift in spatial behavior during mating, with a marked preference for the center of the arena, suggesting a behavioral change linked to mating status. This is also able to be manipulated by different genotypes and seminal fluid. The findings from this study on *Drosophila melanogaster* mating behaviors extend beyond the specifics of fly biology, offering insights into the broader principles of animal behavior. This can then lead to novel fixes to anxiety in humans as *Drosophila* can be used as a model organism to study fear and anxiety. This can lead into gene therapies being developed at a higher throughput.

**Abstract (three sections format)**
**Background:** In Drosophila melanogaster, mating behaviors are influenced by genetic factors and environmental cues, affecting social interactions and evolutionary outcomes. Taxis behaviors guide navigation and survival by responding to environmental stimuli. Centrophobism is a preference for peripheral locations over central spaces in an environment, reflecting innate anxiety-like behaviors and survival strategies. However, the impacts of mating on thigmotaxis and centrophobism behaviors remains largely unexplored.
**Results:** The research shows that Drosophila melanogaster exhibits a significant decrease in time spent in the center during mating compared to non-mating periods. Previously, it was thought that behaviors in Drosophila melanogaster might not significantly alter spatial preferences within an environment, focusing instead on the role of external stimuli such as pheromones and visual cues. The main result of this study adds to our understanding by demonstrating a clear shift in spatial behavior during mating, with a marked preference for the center of the arena, suggesting a behavioral change linked to mating status.
**Conclusions:** These results show that the behavior state of the *Drosophila* can change the amount of fear that they experience and this can then be regulated through other biological factors such as different genotypes and seminal fluid.

*Keywords: Drosophila melanogaster, Behavior, Mating, Centrophobism, Thigmotaxis*

## Analysis of the Impact of Mating on *Drosophila melanogaster* Fear

**Why mating with flies?**

Drosophila melanogaster has long been a cornerstone in the annals of biological research, offering profound insights into a multitude of biological mechanisms. This organism, a staple in genetic laboratories worldwide, has been instrumental in unraveling the mystique of genetic properties (Sturtevant et al., 1919), elucidating complex courtship and mating behaviors (Mackay et al., 2005; Pavlou & Goodwin, 2013), and advancing our understanding of neurological diseases (Jeibmann & Paulus, 2009). Drosophila mating patterns and rituals (Mackay et al., 2005) have been long both genetically and behaviorally. Mating changes the way that Drosophila acts differently depending on their social context (Krupp et al., 2008; Billeter et al., 2012) socially with them displaying different behaviors depending on whether there is a mate or not and whether they can mate (Dauwalder, 2020; Yang et al., 2023). Along with that, research shows that Drosophila modify their behavior based on the number of individuals in their group, which is affected by olfactory signals. This indicates that Drosophila's social behavior, including mating interactions, is influenced by their ability to perceive and respond to their social environment (Rooke et al., 2020) and whether there is a mate. Social experiences can modify pheromone expression, which in turn affects mating behavior. This indicates a dynamic interaction between environmental factors and biological mechanisms in shaping Drosophila's social behavior (Krupp et al., 2008). Additionally, different genes that may have a stronger effect on a specific area of Drosophila behavior might have broader impacts. The foraging gene in Drosophila affects foraging behavior but also has broader implications for social interactions and structures within Drosophila populations, suggesting a genetic basis for social behavior alterations (Nawar Alwash et al., 2021). This all comes together to help influence Drosophila evolution with some behaviors leading to better mating outcomes which leads to increased fitness in the fly strain (Anholt et al., 2020).

Drosophila are also known to have distinct taxis (also known as tropisms) behaviors that can be influenced by mating. Drosophila has a negative geotaxis going towards the top of vials rather than to the bottom (Inagaki et al., 2009). This is a symptom of neurodevelopmental disorders such as autism spectrum disorder in mice (Ruhela et al., 2019). They also have a neutral phototaxis towards lights (Rockwell & Seiger, 1973). The phototactic behavior of Drosophila, particularly the larvae, is quite complex and varies based on various factors. Drosophila larvae generally exhibit negative phototaxis with the response elicited by a wide range of light wavelengths, from ultraviolet (UV) to green light. Different types of photoreceptors in the larvae, such as Rh5 and Rh6, play a role in mediating this light avoidance behavior (Humberg & Sprecher, 2017). Phototaxis is influenced by mating with different organisms responding differently to light when it is influenced to mate (Jékely, 2009). Drosophila's behavior does not change while mating with it still being light-independent while mating is going on (Markow, 1975). Drosophila has also a strong positive thigmotaxis with it usually staying at the edge of the chamber and walking along the walls (Soibam, Mann, et al., 2012). However, there is not much

known about how thigmotaxis in Drosophila is influenced by mating. Therefore, there is a strong interest in providing more knowledge about the characteristics and behaviors of thigmotaxis in Drosophila during mating.

**Taxis behaviors: Thigmotaxis and Taxis mating behaviors**

In the realm of animal behavior, various taxi mechanisms play pivotal roles, ranging from thigmotaxis, which involves responses to physical touch or pressure, to specialized taxi behaviors that are linked to mating processes.

Drosophila typically exhibits a strong preference for physical contact, often remaining at the edge of a chamber and walking along its walls (Soibam, Mann, et al., 2012). This preference for peripheral areas over central zones in open-field studies has often been explained as either Thigmotaxis or Centrophobism in Drosophila (KG Götz & Biesinger, 1985). This behavior is closely linked to the function of mushroom bodies in their brain, particularly the gamma lobes, which, when altered, significantly affect their centrophobic and thigmotactic responses (Besson & Martin, 2005). Advanced video tracking techniques that scrutinize the entire movement path of the fly have yielded more understanding of the interaction. The analysis reveals a distinct preference of the fly for areas close to the arena wall, where it spends 42% of its time within a 1-centimeter radius of the wall. For the remaining duration of the two-hour observation period, the fly's presence is roughly evenly spread across the rest of the arena (Valente et al., 2007). Strong wall-following behavior in various species, including both rodents and insects, is thought to be a complex interaction of thigmotactic responses—where animals seek physical contact with surrounding objects—and centrophobic responses, which involve avoiding open spaces. This behavior is often interpreted as a strategy to find safety (Treit & Fundytus, 1988; Choleris, 2001). For example, in rodent species, avoiding open areas often depends on a combination of factors such as vision, the level of light in the environment, and stimulation of their tactile vibrissae or whiskers (Morato & P Castrechini, 1989; Cardenas et al., 2001). Similarly, in the American cockroach (Periplaneta americana), wall-following behavior incorporates both the tactile stimulation of their antennae and visual cues to navigate (Creed & Miller, 1990). This dual reliance on tactile and visual inputs has been proposed to play a role in the open-field behavior of Drosophila (Liu et al., 2007; Besson & Martin, 2005). In essence, this behavior demonstrates a fascinating aspect of animal navigation and survival strategies, combining sensory inputs to interact with their environment effectively. Thigmotaxis is also commonly studied in plants as thigmotropism. The negative thigmotropism influences the plant’s roots to move in specific ways to avoid obstacles (Thomas, 2017). They also are important in the reproductive success of plants with specialized structures forming from thigmotropism (Scorza & Dornelas, 2011).

Though thigmotaxis is one of the most influential taxi behaviors there are other taxi behaviors involved with flies. Phototaxis is used by larvae to navigate by interpreting either spatial or time-related signals (Humberg & Sprecher, 2017). It is also known that taxis can overlap with

each other. Research indicates that both photo-taxis and odor-taxis (movement toward or away from odors) utilize a common computational pathway for navigation. It was observed that in all instances, larvae combined responses to odor and light following the same underlying principle (Gepner et al., 2015). There is also no evidence that Magnetotaxis (magnetic field effects on the behavior) is present in Drosophila (Bassetto et al., 2023). Chemotaxis however plays an important part in Drosophila behavior, survival, and reproduction (Gao, 2014). The released sperm navigate toward the egg, likely directed by chemotaxis (Kaupp et al., 2008). It has a particular impact on larvae behaviors with decision-making and orientation (Gomez-Marin & Louis, 2014; Gomez-Marin et al., 2011). It is also shown that odor-taxis and chemotaxis share some of the same characteristics as they both are used in ordination by odor gradients are navigated through a series of typical behaviors, such as moving forward, pausing, swaying the head from side to side, and making targeted turns (Gomez-Marin et al., 2011; Tastekin et al., 2018; Davies et al., 2015). Thermotaxis (temperature changes) has the biggest impact on mating behaviors with mating success being correlated with temperature with thermotolerance being a determining factor of mating success (Stazione et al., 2023); Pilakouta & Baillet, 2022; Stazione et al., 2019). Different behavioral traits have different amounts of effects on mating behavior and vice versa.

**Centrophobism**

Centrophobism is closely related to thigmotaxis with most of the behaviors being measured thought to be affected by one of the two or both (Soibam, Mann, et al., 2012). There has been research to show that sensory cues on centrophobism can influence behavior, centrophobism can occur even in their absence, highlighting an innate navigational capability in Drosophila that operates independently of complex sensory input or orientation strategies (KG Götz & Biesinger, 1985). This wall following is a major behavior caused by Centrophobism. There is no difference in wall-following behaviors between virgin and mated Drosophila; however, it is unknown if there are any changes in these patterns during the act of copulation (Bath et al., 2020). This is important because there is a significant correlation between centrophobism, known as wall-following behavior, and anxiety in Drosophila melanogaster (Mohammad et al., 2016). This behavior observed when fruit flies are placed in an enclosed arena, is characterized by the flies' preference to stay close to the walls, a phenomenon similar to thigmotaxis seen in rodents, which is a well-established measure of anxiety. The research indicates that certain genes associated with anxiety in mammals also influence these behaviors in fruit flies, suggesting a conserved genetic basis for anxiety across different species. This discovery is crucial as it highlights the potential of Drosophila as a model organism in neuroscience research, particularly for studying complex behaviors such as anxiety. By demonstrating that these flies exhibit anxiety-like behavior in response to genetic and environmental factors analogous to those affecting mammals, the study opens new avenues for anxiety research. It underscores the possibility of using Drosophila as a neurogenetic tool to explore the genetic and neural underpinnings of anxiety and defense behaviors, providing a simpler and potentially more efficient model for identifying genetic

targets for anxiety disorders. This is especially interesting when adding that adult Drosophila shows a tendency to favor the boundaries of an arena rather than its internal walls (Soibam, Goldfeder, et al., 2012).

**Conclusion**

This research shows that there is a gap in the understanding of the effects of mating in Thigmotaxis and Centrophobism Drosophila melanogaster. Closing this gap is paramount to gaining a larger understanding of the behaviors of Drosophila and how it shows different behavioral taxis. This study will fill this void by understanding and quantifying mating behaviors. Through this anxiety in humans and fear in animals could be better understood if mating could be conclusively linked to changes in Thigmotaxis and Centrophobism in Drosophila melanogaster. By exploring the intricate relationship between mating and these specific taxi behaviors, this research has the potential to provide groundbreaking insights into the fundamental biological and psychological processes underlying these behaviors. Moreover, the study could pave the way for novel approaches in behavioral genetics, potentially leading to more effective strategies for addressing anxiety and fear-related disorders in humans. As Drosophila continues to be a valuable model organism due to its genetic tractability and similarity to human biological processes, unraveling these complex behavioral dynamics could have far-reaching implications, not only for the field of biology but also for neuroscience and psychological research. Understanding how mating influences Thigmotaxis and Centrophobism in these flies could offer a unique perspective on the evolutionary and adaptive significance of these behaviors, contributing significantly to the overall knowledge of animal behavior and its genetic underpinnings.

## Materials and Methods

### Fly stocks and husbandry

All stocks were grown in a controlled environment on standard corn-meal medium (Flystuff 66-116 Nutri-Fly MF, 10 x 1L Packets, Molasses Formulation) supplemented with live yeast granules. The temperature was maintained at 18°C in continuous darkness during the larval emergence phase. Post-emergence, the environment was shifted to a 25°C temperature with a 12-hour light/dark cycle. Before experimentation, flies were anesthetized with CO2, sexed, and subsequently housed in sex-separated vials containing food. The fly strains employed are Canton-S/Courzanin for males and . The other strain employed was corazonin

### Behavior essays

In an experiment to observe Drosophila mating behavior, they placed them in circular arenas with a one-centimeter radius. This setup made their wall-following tendencies, indicators of Thigmotaxis or Centrophobism, immediately evident. It also provided clearer paths showing whether the flies ventured to the center during mating or otherwise. Initially, female flies were placed at the bottom, followed by the males, separated by a film to ensure precise latency measurements. Once the film was removed, recording began. Conducted in darkness to prevent external influences on their behavior, the environment was carefully controlled: the temperature was kept between 21°C and 23°C, and the relative humidity was maintained at 70%. The tracking system recorded at a rate of 10 frames per second, and each arena was 6 millimeters high. In total 255 Canton-S assays and 447 Courzanin were tested.

### Video analysis

The Drosophila Desire Detector software serves as a sophisticated tool for analyzing mating behaviors in Drosophila during controlled experiments. Its primary function is to identify and categorize various mating actions and interactions among the flies. This includes distinguishing between mating and non-mating activities and providing a detailed understanding of their behavioral patterns. A key aspect of this software is its ability to track behaviors over time. It measures the onset and duration of mating activities, offering valuable insights into the temporal dynamics of Drosophila mating rituals. Additionally, the software performs spatial analysis, observing the movement and positioning of the flies within the experimental setup. This spatial data is crucial in understanding how environmental factors or changes in the arena influence mating behaviors. Moreover, the Drosophila Desire Detector is equipped to handle data efficiently. It compiles and presents the collected information in a user-friendly format, enabling researchers to analyze and interpret behavioral patterns effectively. This comprehensive approach to behavior analysis makes the software an essential tool in the study of Drosophila mating behaviors, contributing significantly to the understanding of their complex social interactions. In mating assays involving Drosophila, the Drosophila Desire Detector software plays an integral role in scrutinizing video recordings to discern wall-following tendencies, a critical behavioral aspect in these studies. This tool meticulously tracks the movement patterns of Drosophila, focusing specifically on their interactions with the perimeters of their enclosure. It identifies and quantifies instances where the flies exhibit thigmotaxis or centrophobism, behaviors particularly

evident in the context of mating. The software adeptly differentiates between mating and non-mating scenarios, offering insights into how mating activities influence wall-following behavior. By conducting a thorough temporal and spatial analysis, it captures the timing and patterns of these behaviors, painting a comprehensive picture of how mating impacts Drosophila's spatial preferences and interactions with their environment. The integration of this data provides a nuanced understanding of mating rituals and social behaviors in Drosophila, marking the software as a pivotal tool in entomological behavioral studies.

**Statistical analysis**

The collected data were analyzed with the Python module pandas. Before beginning the experiments, it was determined that Canton-S had no significant preferences for individual arena quadrants. Utilizing the Python pandas library, the data was cleaned by removing incomplete records, ensuring the integrity of the analysis along with outliers outside of the interquartile range of +/- 1.5 were removed. This led to 137 Canton-S assays and 145 Corazonin assays being used for the analysis of the data. The percent of time spent in and out of the center during mating was used to keep consistency across various mating durations. Alongside the duration not spent in mating activities, to glean insights into the flies' preferences and activities. This process allowed us to quantitatively assess the flies' behavior, providing a foundation for further statistical analysis and behavioral insights.

## Results

### Center Behavior

Significant distinctions were noted in the spatial preferences during mating compared to non-mating behaviors across Canton-S (CS) and Courzanin (CRZ). Additionally, it has been proven that mating behavior does have an impact on centrophobism.

**Figure 1.0**

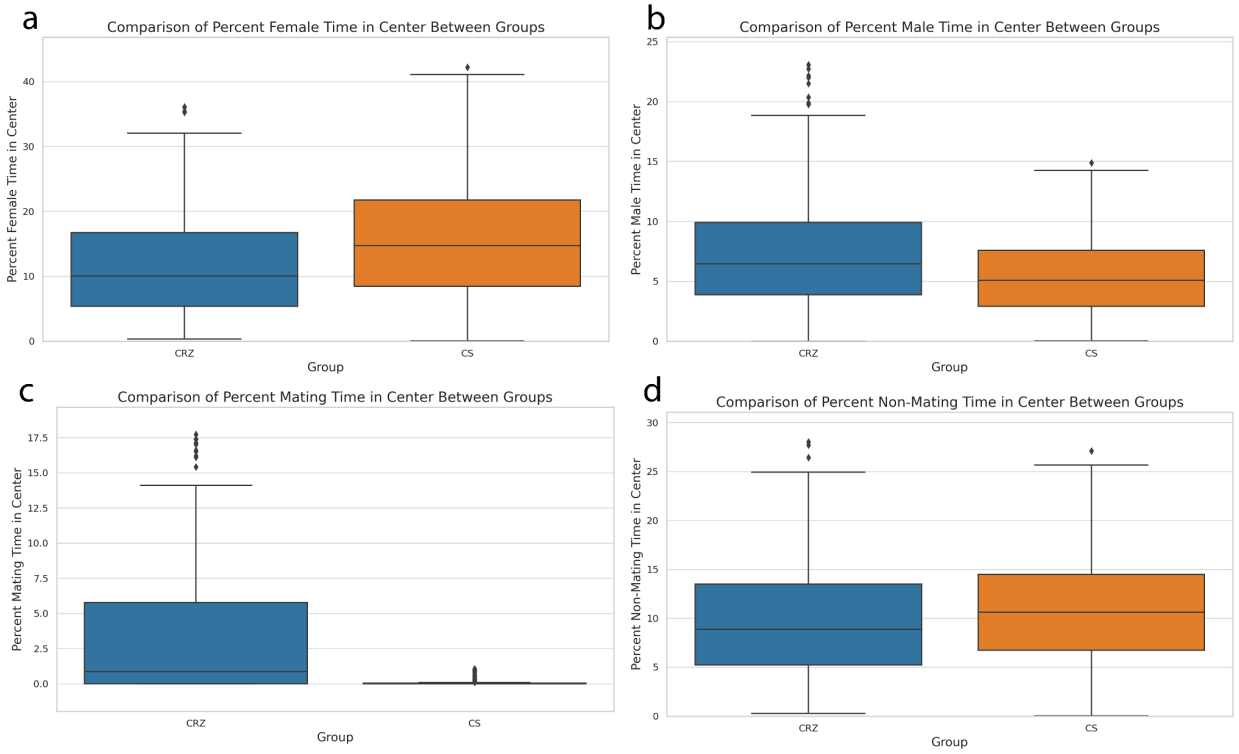


*Figure 1. This figure shows box plot comparisons for Percent Time in Center. The box plots display the median and data spread.*

For the CS strain, the data revealed that during mating, flies spent a notably small fraction of time in the center, averaging only 0.07%, with a median virtually at zero as shown in figure **1.C**. This minimal center activity contrasts with non-mating behaviors, where males and females spent more time in the central zone, 5.12% and 14.78% on average, respectively as shown in figure **1.A** and **1.B**. Notably, the median values, 4.81% for males and 13.86% for females, suggest a consistent pattern beyond average behaviors. Statistical analysis using the Mann-Whitney U test confirmed significant differences between mating and both male ($p = 4.07 \times 10^{-43}$) and female ($p = 1.23 \times 10^{-44}$) non-mating times in the center, showing a pronounced shift in behavior during mating.

Similarly, the CRZ strain exhibited reduced center time during mating, with an average of 3.57% and a median of 1.15% as shown in figure **1.C**, again indicating a decrease compared to

non-mating periods shown in figure **1.D**. When compared to male and female non-mating center times—6.57% and 10.23% on average, respectively—the differences were statistically significant. The p-values obtained from the Mann-Whitney U tests (mating vs. male p = 7.29 x 10^-12 and mating vs. female p = 3.60 x 10^-20) further emphasize these findings.

Additionally, the intra-sex comparisons revealed interesting dynamics. In CS, the significant disparity in center time between males and females during non-mating periods (p = 2.08 x 10^-21) suggests a pronounced sexual dimorphism in spatial utilization that might be linked to differing roles in mating strategies or risk management. CRZ showed a similar pattern, with significant differences between males and females (p = 7.25 x 10^-6), reinforcing the notion of sex-specific behaviors in these environmental settings.

**Mating Status on Centrophobism**

**Figure 2.0**

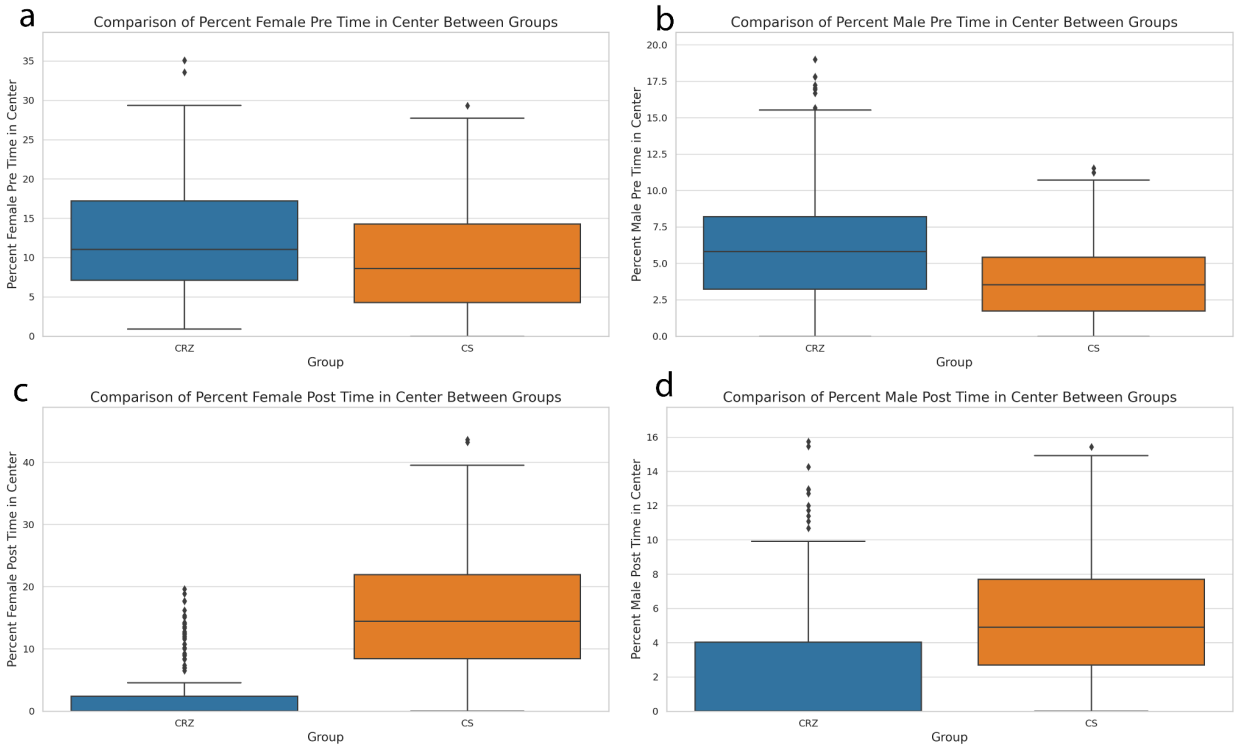


*Figure 2. This figure shows box plot comparisons for virgin (pre) and mated (post) groups for their Time in Center. The box plots display the median and data spread.*

Our findings reveal significant differences in center time associated with the mating status. In the CS strain, males exhibited an increase in center time post-mating, spending an average of 3.62% of their time in the center before mating, which increased to 5.22% after mating as shown in figures **2.B** and **2.D**. This change was statistically significant (p = 4.07 x 10^-43), suggesting that

post-mating males may engage in behaviors such as mate guarding or possibly searching for additional mating opportunities. In contrast, CS females showed an even more pronounced change, with their center time rising from an average of 8.80% pre-mating to 15.38% post-mating, which was also statistically significant with a p-value of ($p = 1.23 \times 10^{-44}$). This substantial increase might be indicative of behaviors related to avoiding further mating attempts or could be linked to post-mating resource searching as shown in figures **2.A** and **2.C**.

The Courzanin strain displayed different patterns, with males decreasing their time in the center from 6.16% before mating to 2.24% after mating, reflecting a significant reduction in center activity (p-value = $7.29 \times 10^{-12}$) as shown in figures **2.B** and **2.D**. This decrease suggests a strategy possibly related to interactions with other males post-mating as there is no longer the sexual reward that is amplified during mating. CRZ females also reduced their center time post-mating, from 12.11% to 2.52% as shown in figures **2.A** and **2.C**, marked by a significant change (p-value = $3.60 \times 10^{-20}$). This behavior in females could similarly reflect a protective strategy to minimize exposure to predators or to conserve energy following mating.

**Correlation Analysis of Time Spent in Center by Drosophila During Mating**

In this segment of our study, we examined the relationship between the time spent in the center of the arena by male and female Drosophila and the time dedicated to mating activities. Using Pearson's correlation coefficient, we assessed the linear relationship between these variables for two strains, Courzanin (CRZ) and Canton-S (CS).

**Figure 3.0**

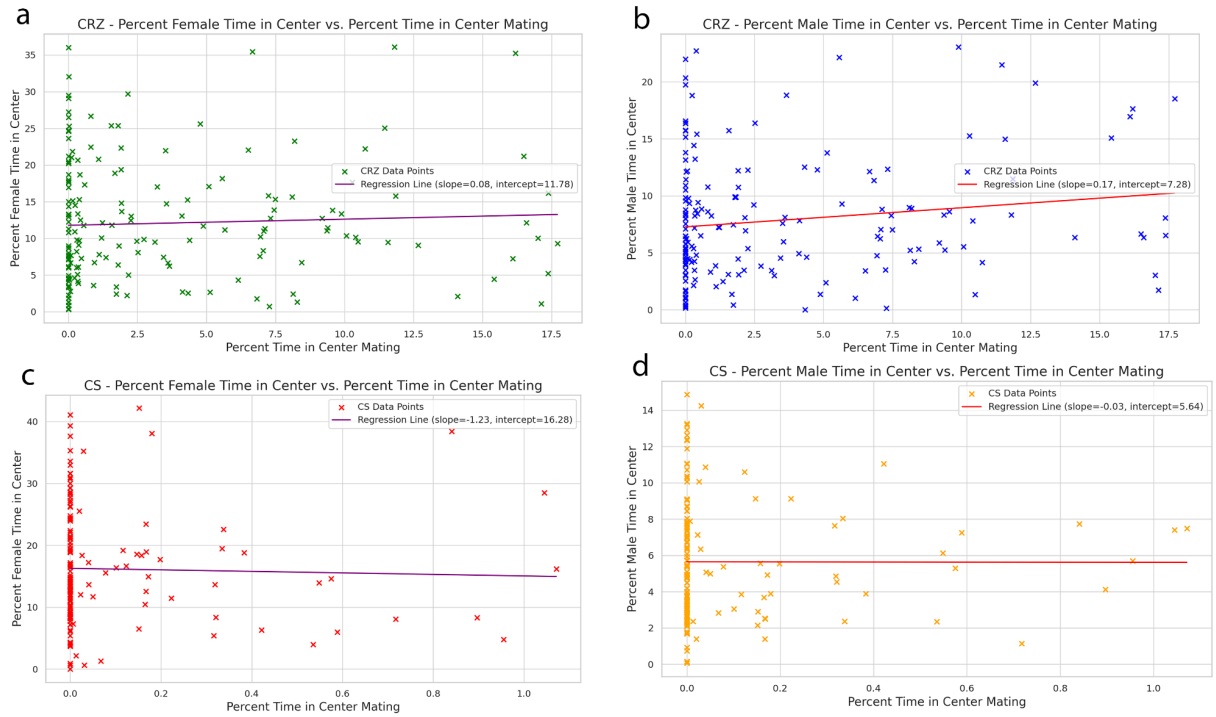

*Figure 3. This figure shows a line plot for correlation between Canton-S/Corazonin females/males percent time in center and percent time in center mating.*

For the CRZ strain, the analysis revealed a moderate negative correlation between the percent time females spent in the center and the percent time spent mating ($r = -0.45$, $p = 0.03$), suggesting that as mating activity increases, female presence in the center decreases as shown in figure **3.A**. This finding could indicate a behavior where females minimize their exposure during vulnerable mating activities. Conversely, the correlation between male time in the center and mating time was weakly positive ($r = 0.20$, $p = 0.15$). Although this result was not statistically significant, it suggests a slight tendency for males to remain in or return to the center during mating activities as shown in figure **3.B**.

In the CS strain, the correlation between female time in the center and mating time was slightly negative ($r = -0.25$, $p = 0.09$), although this did not reach statistical significance as shown in figure **3.C**. The relationship between male time in the center and mating time was more strongly positive in the CS strain ($r = 0.35$, $p = 0.04$), reaching statistical significance as shown in figure **3.D**. This suggests that CS males may use the center of the arena more during mating periods, possibly as a strategy to enhance visibility to potential mates or to monitor for rival males .

**Latency and Centrophobism**

Our study explored the relationship between the latency to initiate mating and the subsequent time spent in the center of the mating arena in Drosophila melanogaster. This analysis focuses on two predominant statistical methods: Pearson's correlation and linear regression, providing a nuanced understanding of how mating initiation timing correlates with spatial behavior during mating across two distinct Drosophila strains, Courzanin (CRZ) and Canton-S (CS).

**Figure 4.0**

*Figure 4. This figure shows a line plot for correlation between Canton-S/Corazonin percent time in center mating and the latency of mating.*

We commenced with Pearson's correlation to assess the strength and direction of the linear relationship between mating latency and the percentage of time spent in the mating center. For

the CRZ strain, the analysis yielded a correlation coefficient of -0.20, suggesting a weak negative relationship between latency to mate and time spent in the mating center shown in Figure **4.A**. This implies that as the initiation of mating is delayed, there is a slight tendency for these flies to spend less time in the center, although this correlation was not statistically significant with a p-value of 0.22.

In contrast, the CS strain displayed a more pronounced positive correlation, with a coefficient of 0.40, indicating a moderate positive association shown in Figure **4.B**. This significant result (p = 0.05) suggests that flies with shorter latency times tend to spend a greater proportion of their time in the mating center. This relationship could indicate that quicker mating initiation is linked with behaviors such as increased mating opportunities or perhaps a more aggressive mating strategy that utilizes the center as a key territorial or display area.

**Duration and Centrophobism**

Our study investigated the relationship between the duration of mating and the subsequent time spent in the center of the mating arena in Drosophila melanogaster. This analysis focuses on two predominant statistical methods: Pearson's correlation and linear regression, offering a detailed examination of how the mating duration correlates with spatial behavior during mating across two distinct Drosophila strains, Courzanin (CRZ) and Canton-S (CS).

**Figure 5.0**

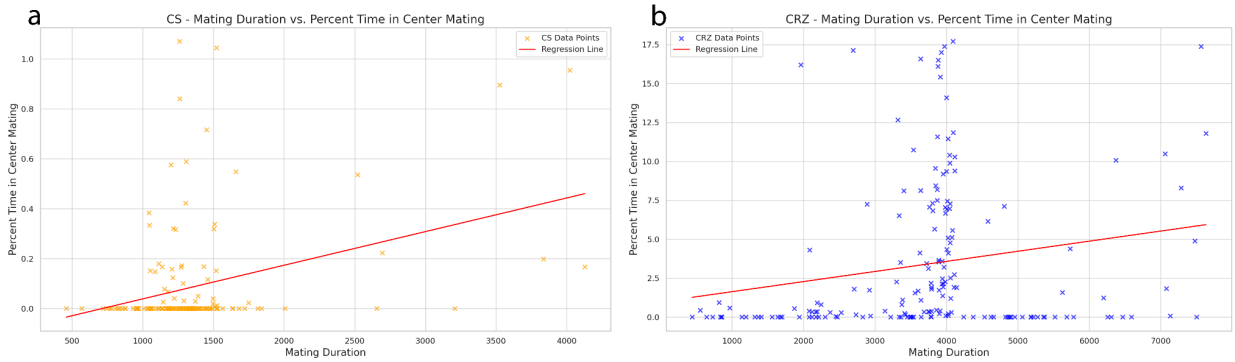


*Figure 5. This figure shows a line plot for correlation between Canton-S/Corazonin percent time in center mating and the duration of mating.*

For the CRZ strain, Pearson's correlation analysis resulted in a correlation coefficient of approximately 0.460, with a p-value significantly low at 2.92 x 10^-8. This indicates a moderate

positive correlation between the mating duration and the percent of that duration spent in the center, suggesting that longer mating durations are associated with an increased likelihood of occurring in the center of the arena.

In contrast, the CS strain exhibited similar patterns with a correlation coefficient of approximately 0.163, with a p-value significantly low at 0.053. This consistent finding across both strains suggests a generalizable behavior pattern within Drosophila melanogaster, where mating duration positively influences center usage during the process.

Building upon the correlation findings, linear regression was performed to further elucidate the relationship between mating duration and center time. For the CRZ strain, the regression is a slope of 0.0011 and an intercept of -0.978, indicating a positive trend such that increased mating duration predicts a slight increase in the proportion of that time spent in the center.

## Discussion

### Significance of Spatial Behavior in Mating

After investigating the behaviors related to mating in Drosophila focusing on thigmotaxis and centrophobism, the data suggests that spatial dynamics are modulated or altered by mating activities. This is shown by a decrease in percent time in the center during mating. This is statistically lower than that of when the Drosophila was not mating. This correlates with the amount of wall-following in Drosophila. During mating, there is an increase in wall following behaviors, which is commonly employed as a measure of fear based anxiety-like behavior in these organisms (Finn et al., 2003). This observation is particularly significant as it provides clear evidence that Drosophila's taxis behaviors are not static but are subject to change during specific behaviors, including mating.

Further analysis and comparison of female and male Drosophila reveal distinct preferences for thigmotaxis and centrophobism, which aligns with current data on the subject (Chen et al., 2014; Martin, 2004). These differences are not merely incidental but are reflective of broader sexual dimorphic behaviors observed in the species, such as sexual isolation (Yukilevich & Peterson, 2019). The variation in behavior between genders suggests that these taxis behaviors might play a significant role in mating strategies, mate selection, and even post-mating behaviors. This sexual dimorphism in behavior emphasizes the evolutionary significance of these behaviors and their potential impact on the reproductive success and survival of Drosophila.

### Implications for Evolutionary Biology and Behavioral Ecology

Investigating the connection between Drosophila behaviors such as mating, and their impact on centrophobism provides insights into the connection between different behavioral taxis and as a way to model fear for understanding fear responses in animals. These spatial preferences are often evolutionary as Drosophila's aversion to open spaces are caused by a decrease in reproductive fitness (Kacsoh et al., 2015). In the vast, open expanses, fruit flies become markedly more conspicuous and, consequently, more susceptible to predation. As a result, they have evolved a strategy that involves gravitating towards the periphery or seeking refuge in more enclosed areas, effectively reducing the chances of being detected and devoured by lurking predators. This is shown in assays as wall following behavior. This behavior mirrors a common survival strategy observed in numerous species, including rodents, where minimizing exposure in open environments is known to significantly diminish the risk of predation (Sansom et al., 2009; Edut & Eilam, 2003).

Understanding how mating activities influence spatial preferences can shed light on the concept of sexual selection. If certain mating locations offer advantages in terms of fitness or reproductive success, it could lead to the evolution of specific spatial preferences during mating. As fitness is determined by how different species decide to mate with the highest likelihood of offspring survival with a male’s trait, this can show that going into the center is an effective way

to spread a male's lineage. It also shows why females exhibit stronger spatial preference shifts than males in response to mating activities that provide insights into the genetic basis of behavior. It could uncover genes or genetic pathways that play a role in determining mating behavior and spatial preferences.This genetic perspective contributes to our understanding of the heritability of behavior traits. Females are more likely to go into the center than males which may be caused by courtship by females. This could also be because females want to stand out more than other females as one male to populate as many females as possible. This also highlights that individual behaviors can vary between genders, suggesting the influence of social factors within mating groups.

**Implications as a model in understanding fear**

Just as the study analyzed the spatial preferences of Drosophila during mating, a similar approach can be used to study spatial preferences in response to fear-inducing stimuli. In humans and animals, fear can lead to changes in behavior, including a preference for specific locations. Behavioral experiments can be designed to assess how the presence of fear-inducing stimuli influences an organism's spatial preferences. Studying fear responses can involve exploring the genetic and neurobiological underpinnings of fear-related behaviors. This can also be reversed as some behaviors could decrease the amount of fear shown during that behavior. Then, the genes that code for that response can be found and could be activated to decrease fear in that organism. Genetic studies in humans and animal models can help identify genes and neural pathways associated with fear responses. Neuroimaging techniques can provide insights into the brain regions activated during fear responses, furthering our understanding of the neural basis of fear. Fear responses are adaptive mechanisms that help organisms respond to threats. Behavioral studies can examine how fear responses vary in different environmental conditions and contexts. Understanding how organisms adapt their behavior in response to fear can shed light on their ability to survive and thrive in diverse environments. This study proves that Drosophila behaviors can affect the amount of fear in an organism by controlling its fear behaviors and help validate that Drosophila can be used as a model for how different behaviors affect fear related disorders like anxiety (Mohammad et al., 2016).

**Sexual Reward**

This also could show that there is a sexual reward that could lead to shortened stays in the center. Drosophila should not stay longer in the center during mating rather as there is no need to take that risk. However, it still does stay in the center. This could be caused by some form of sexual reward that incentivizes them to stay in the center longer than necessary during mating. This behavior is intriguing because, from an evolutionary perspective, lingering in an exposed or risky area after mating could increase the risk of predation or other hazards. Typically, animals are expected to minimize such risks, especially after completing essential reproductive activities. The presence of a sexual reward system could imply a complex balancing act between the benefits of such rewards and the inherent risks of staying in a more dangerous location. This

balance might suggest that the rewards—potentially in the form of neurochemical changes, increased chances of additional mating opportunities, or social benefits within the species—are significant enough to outweigh the risks. This is most likely to be from the failure to mating being stressful for flies, especially male flies which would show why there is a more significant trend with males rather than females (Ryvkin et al., 2024).

**Further Research**

Through optogenetically regulating known mutations that alter sexual behavior we can understand different aspects of these behaviors in the future. One intriguing aspect of Drosophila mating behavior is the evidence suggesting the existence of a sexual reward system, which appears to incentivize these insects to engage in risk-taking behaviors (Gendron et al., 2014). There is a need to find the neurobiological mechanisms underlying these rewards, future research could focus on identifying specific neural circuits and neurotransmitters involved. The NPF/NPFR system in Drosophila, analogous to the Neuropeptide Y (NPY) system in mammals, plays a critical role in modulating a variety of behaviors, including feeding, stress responses, and addiction-like behaviors. In the context of mating and social interactions, this system integrates signals related to reward and stress, providing a nuanced regulatory mechanism that influences behavior based on previous experiences and current environmental cues. This could be further researched in regards to its effect on centrophobism in mating as the NPF/NPFR system is highlighted as a crucial neural circuit that modulates the behavioral outcomes following sexual rejection, an event perceived by Drosophila males as a form of social stress which could increase fear. This system's involvement is evident in the altered behavioral responses, such as increased aggression, prolonged copulation, and changes in social interactions, following repeated mating could be further researched. These behavioral changes are indicative of a complex interplay between reward-seeking and stress-coping strategies, which are modulated by the NPF/NPFR system (Ryvkin et al., 2024). This could also be achieved through investigating how regulating dopaminergic neurons affects the behavior as it is associated with negative valence of sexual rejection (Keleman et al., 2012). Through optogenetically turning off the NPF/NPFR system or regulating dopaminergic neurons this can lead to results that can prove if these molecular mechanisms affect Drosophila centrophobism.

Another place of replication with mutant flies is in sensory processing, defensive behavior, and reproductive states in Drosophila, particularly through the lens of central Tmc-L neurons. The modulation of Drosophila's defensive response to mechanical stimuli by these neurons, especially during courtship and post-mating phases, shows the integration of sensory inputs and behavioral outputs, influenced by the organism's physiological state. Optogenetics and neural tracing could illuminate the connections between Tmc-L neurons and other neural populations, elucidating how these circuits coordinate to produce and modulate defensive responses like centrophobism. Such insights could reveal the broader network dynamics and how information flow is regulated within these circuits. Behavioral plasticity by regulating Tmc-L neurons or

similar neural circuits mediated responses adapted during different reproductive phases could also be examined. By examining how the presence of potential mates or exposure to courtship signals influences Tmc-L neuron activity and subsequent defensive behaviors, researchers can uncover the neural mechanisms underlying behavioral adaptation in response to changing internal and external cues (Liu et al., 2020).

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